

Carbon Now!

A Carbon lifecycle calculator
for everyone

Team 8

**Architects
Managers
No Coding
Experience
No AI Experience**

**NEURAL
NETWORKS**
for
babies

The Team



A photograph of a modern office interior. Three people are standing and talking. On the left is a woman with dark hair in a ponytail, wearing a dark blue polo shirt. In the center is an older man with grey hair and glasses, wearing a dark grey polo shirt. On the right is a younger man with short dark hair, wearing a dark blue t-shirt. They are all looking at each other. The background shows office shelves with plants, a large window, and a ceiling-mounted air conditioner. A red speech bubble points from the woman to the text.

Can you figure
out the ballpark
and get back to
me?

The Situation

A group of people are working in a modern, plant-filled office. They are seated at a long wooden table, using laptops. The office has large windows, hanging plants, and warm lighting. The scene is a collaborative workspace.

Can we learn from scratch and build a tool in an area with limited knowledge?

The Challenge

The Task

Create a custom GPT dedicated to on-demand calculations of any building's full carbon life cycle

The Tool

Large Language Models (ChatGPT)

The Challenge

Use no code tools to train LLM models & create new machine-readable database

There are many tools dedicated for Specialist
But none for designers or any tool that allows
for a quick overview, rapidly

Demo Calculator

Calculate a localized estimate (for Singapore) for embodied carbon emissions, just by entering a building material and quantity!

Category

Sealant

▼

Component

Epoxide Resin

▼

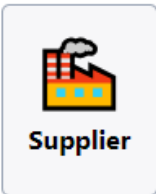
Quantity

Units

kg

▼

☒ I have transport information



Supplier



International Road Vehicle

None

▼

International Road Distance

km

Postal Code of Factory

123456



Maritime Vehicle

None

▼

Country of International

Port

None

▼

Postal Code of International

Lo

Lo

Po

A1 - A3 emissions

0.00 kgCO2eq

A4 emissions

0.00 kgCO2eq

Total emissions

0.00 kgCO2eq

Sieve Through Data
ICE, Energy Usage,
Building Data



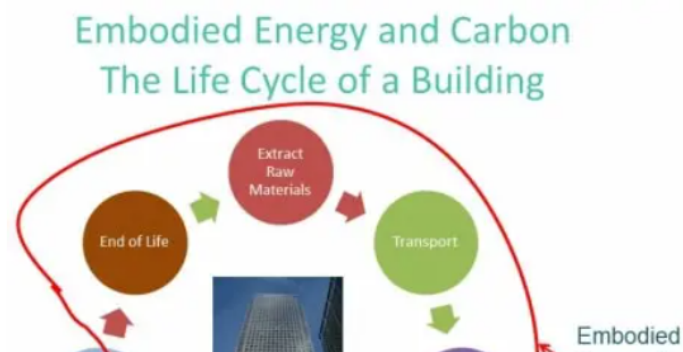
Instant Estimation
Quick estimation of
full-cycle carbon
footprint

Embodied Carbon - The ICE Database

Embodied carbon comes from the consumption of embodied energy consumed to extract, refine, process, transport and fabricate a material or product (including buildings). It is often measured from cradle to (factory) gate, cradle to site (of use), or cradle to grave (end of life). The embodied carbon footprint is therefore the amount of carbon (CO₂ or CO_{2e} emission) to produce a material.

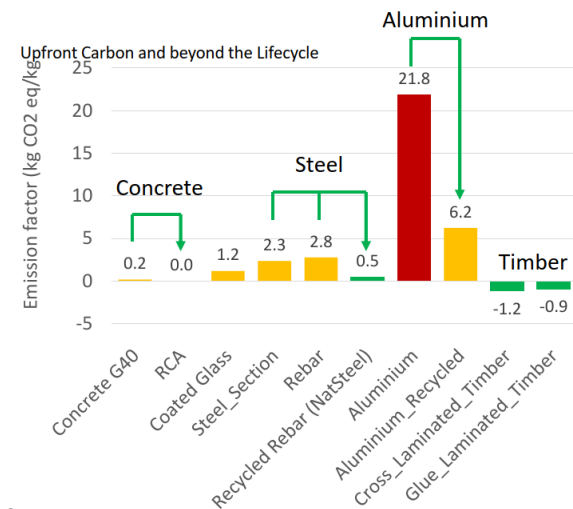
It is a topic of rising importance. In fact, it is normally possible to reduce the embodied energy and carbon of a building or construction project by 10-20% without adding to the build cost. What's more, embodied carbon is often 20-50%, or so, of the whole life energy and carbon of a building, i.e. when operational carbon emissions are considered.

The embodied carbon life cycle of a building can be expressed on a single diagram, as below.



Current status – Embodied Carbon of Common Construction Materials

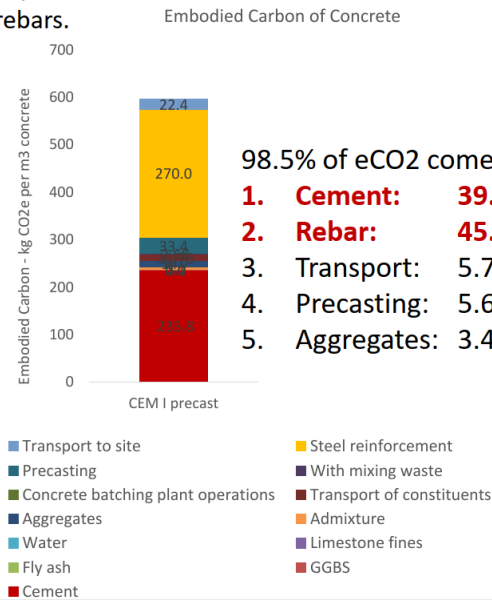
While timber products show negative eCO2 in the carbon calculators, the expected application is still limited in SG.



Source:

1. NUS-ESI (2022), Building Embodied Carbon Calculator (BECC)
2. BRE (2021), Environmental Production Declaration (EPD) for 1 tonne steel reinforcing bar/coil – NatSteel Holdings Pte Ltd.

Reinforced concrete structures will still dominate. The key to decarbonize construction is within concrete and rebars.



98.5% of eCO₂ comes from:

- | | | |
|----|-------------|-------|
| 1. | Cement: | 39.5% |
| 2. | Rebar: | 45.2% |
| 3. | Transport: | 5.7% |
| 4. | Precasting: | 5.6% |
| 5. | Aggregates: | 3.4% |

C.Scale calculates whole life carbon emissions

C.Scale integrates embodied, operational, and landscape carbon emission assessment into a single model. By taking a 'whole carbon' view, C.Scale prevents burden shifting and ensures that a project has the information necessary to target the most impactful carbon reductions.

C.Scale uses GWP-100 characterization factors.

Calculating Embodied Carbon [↗](#)

For each contributor i , embodied emissions are assessed with the following expression:

$$Assessed\ emissions_i = A * x_i * c_i$$

Where A is the total building area, x_i is the quantity of the contributor i per building area, and c_i is the carbon intensity per unit of the contributor i .

For example, a 10,000 square foot building may use 4 pounds of reinforcing steel per square foot of floor area, and the reinforcing steel may have a carbon intensity of 500 grams (0.5 kilograms) of carbon dioxide-equivalent emissions per pound of steel (values for illustrative purposes only). Taking the product of these three hypothetical quantities yields the contribution of reinforcing steel to that building's embodied carbon emission:

$$10,000 \text{ sf} * 4 \frac{\text{lbs rebar}}{\text{sf}} * 0.5 \frac{\text{kg CO}_2\text{e}}{\text{lb rebar}} = 20,000 \text{ kg CO}_2\text{e}$$

Downloaded from <http://ajphaphysiol.phapublications.org/> on September 11, 2015

Scholar GPT ▾

You

Search for articles which talk about embodied carbon calculation for buildings estimation

Scholar GPT

Talked to actions.sider.ai

Here are some relevant articles on the estimation and calculation of embodied carbon in buildings:

1. Estimation and Minimization of Embodied Carbon of Buildings: A Review

- **Authors:** A. Akbarnezhad, J. Xiao
- **Publication:** Buildings, 2017
- This review article discusses methods for estimating and minimizing embodied carbon from the construction phase to the end of life of buildings.
- [Read more](#)

2. An Analytical Method for Evaluating and Visualizing Embodied Carbon

Emissions of Buildings

- **Authors:** E. Resch, C. Lausset, H. Brattebø, I. Andresen
- **Publication:** Building and Environment, 2020
- The study presents a method for evaluating and visualizing embodied carbon emissions, which can be applied to different building types to gain insights into emission profiles.

**We searched
(with ChatGPT help)**

OU

You

calculate the whole life carbon of marina bay sands building in singapore



ChatGPT

Calculating the whole life carbon of a complex building like Marina Bay Sands in Singapore involves a detailed analysis encompassing several components, including the embodied carbon of construction materials, operational carbon from energy use, and the potential end-of-life impacts. However, as an AI developed by OpenAI, I don't have access to specific proprietary or detailed data about the building's materials, construction processes, and operational specifics required to provide an exact calculation.

Here's a general approach on how such a calculation could be conducted, assuming one has access to the necessary data:

1. Data Collection:

- Gather detailed information on all materials used in the construction, including quantities and types of concrete, steel, glass, etc.
- Obtain data on the construction process, including equipment used and duration of construction activities.
- Collect annual energy consumption data for heating, cooling, lighting, and other building operations.
- Estimate the lifespan of the building and the maintenance/renovation activities expected during its life.

2. Embodied Carbon Calculation:

- Use databases like the ICE (Inventory of Carbon & Energy) Database to find the embodied

Current GPT models are not accurate & require more datasets



Message ChatGPT...



Track your carbon footprint

An average Singaporean emits around **730** kg of CO₂ each month. This means that each person would require around **36** rain trees to absorb their carbon emissions!

Before calculating your footprint, take a quick guess!

How many rain trees do you think it takes to absorb your carbon emissions every month?

[Optional]

Get Started

Download the SP App now to keep track of your CO₂



The Process

Step	Task	Process	Database
1	Get Building Data	Based on user input of name or address, or site area, searches database of buildings for match. Extracts key area, type, EUI information. If unavailable, search internet for information and correlate.	Building Energy Performance Database. Internet scrapped database.
2	Calculate Embodied Carbon	Based on building type, given building structure and envelope type, assume building envelope type if not given. Use formula from database to Calculate Embodied Carbon.	C.Scale formula, ICE DB.
3	Calculate Operational Carbon	Based on building database, use EUI of year against EMA dataset of year. If unavailable, search for similar types and estimates.	SES 2023 / EMA data of Carbon / kgco2 per Kwh by year
4	Calculate Carbon Sequestration	Sum embodied and operational carbon to determine total emissions. Calculate the number of trees required to offset these emissions over the building's current lifetime.	Tree offset database from tabulated research

Building Name	building address	building type	building size	green mar	greenmar	green mark stat	GFA	2018energ	2019energ	voluntarydisc
1 NEXUS @ ONE-NORTH	3 FUSIONOPOLIS LINK, SINGAPORE 138543	Commercial Building	Large	2019	Platinum	Yes	25,510	25	23	Y
2 BCA ACADEMY OF THE BUILT ENVIRONMENT	200 BRADDELL ROAD, SINGAPORE 579700	Commercial Building	Large	2017	Platinum	Yes	53,422	41	39	Y
3 N/A	N/A	Commercial Building	Large	N/A	N/A	Yes	N/A	54	50	N
4 N/A	N/A	Commercial Building	Large	N/A	N/A	No	N/A	70	74	N
5 N/A	51 GRANGE ROAD, SINGAPORE 249564	Commercial Building	Large	2017	Certified	Yes	20,916	73	74	Y
6 JURONG TOWN HALL	9 JURONG TOWN HALL ROAD, SINGAPORE 609431	Commercial Building	Large	2017	Platinum	Yes	26,248	77	77	Y
7 GREAT EASTERN @ CHANGI	200 CHANGI ROAD, SINGAPORE 419734	Commercial Building	Large	N/A	N/A	No	24,707	103	77	Y
8 MITSUBISHI ELECTRIC BUILDING	307 ALEXANDRA ROAD, SINGAPORE 159943	Commercial Building	Large	2014	Platinum	Yes	16,654	95	80	Y
9 N/A	1 DEFU AVENUE 1, SINGAPORE 539540	Commercial Building	Large	N/A	N/A	No	19,516	84	82	Y
10 N/A	N/A	Commercial Building	Large	N/A	N/A	Yes	N/A	95	89	N
11 PLUS	20 CECIL STREET, SINGAPORE 049705	Commercial Building	Large	N/A	N/A	No	32,383	66	89	Y
12 N/A	501 OLD CHOA CHU KANG ROAD, SINGAPORE 698928	Commercial Building	Large	2019	Platinum	Yes	137,488	120	90	Y
13 SCDF JALAN BAHAR CAMP	91 JALAN BAHAR, SINGAPORE 649735	Commercial Building	Large	N/A	N/A	No	34,622	88	91	Y
14 EIGHTRIUM @ CHANGI BUSINESS PARK	15A CHANGI BUSINESS PARK CENTRAL 1, SINGAPORE 486035	Commercial Building	Large	N/A	N/A	No	19,876	116	92	Y
15 N/A	18 NAPIER ROAD, SINGAPORE 258510	Commercial Building	Large	2014	Platinum	Yes	15,211	99	93	Y
16 ICON@IBP	3A INTERNATIONAL BUSINESS PARK, SINGAPORE 609935	Commercial Building	Large	2009	GoldPlus	Yes	41,135	102	96	Y
17 N/A	1 KAY SIANG ROAD, SINGAPORE 248922	Commercial Building	Large	N/A	N/A	No	15,091	98	98	Y
18 AKZONOBEL HOUSE	3 CHANGI BUSINESS PARK VISTA, SINGAPORE 486051	Commercial Building	Large	2019	Gold	Yes	19,225	102	98	Y
19 N/A	N/A	Commercial Building	Large	N/A	N/A	Yes	N/A	124	110	N
20 TOWER FIFTEEN	15 HOE CHIANG ROAD, SINGAPORE 089316	Commercial Building	Large	N/A	N/A	No	23,546	144	111	Y
21 HIGH STREET CENTRE	1 NORTH BRIDGE ROAD, SINGAPORE 179094	Commercial Building	Large	2018	Platinum	Yes	43,000	165	111	Y
22 N/A	31 INTERNATIONAL BUSINESS PARK, SINGAPORE 609921	Commercial Building	Large	N/A	N/A	No	61,720	118	112	Y
23 GERMAN CENTRE	25 INTERNATIONAL BUSINESS PARK, SINGAPORE 609916	Commercial Building	Large	2015	Platinum	Yes	28,421	111	113	Y
24 TONG ENG BUILDING	101 CECIL STREET, SINGAPORE 069533	Commercial Building	Large	2013	Platinum	Yes	22,930	113	114	Y
25 KEPPEL TOWERS	10 HOE CHIANG ROAD, SINGAPORE 089315	Commercial Building	Large	N/A	N/A	No	58,609	156	114	Y
26 ENVIRONMENT BUILDING	40 SCOTTS ROAD, SINGAPORE 228231	Commercial Building	Large	2018	Platinum	Yes	49,021	122	115	Y
27 NORDIC EUROPEAN CENTRE	3 INTERNATIONAL BUSINESS PARK, SINGAPORE 609927	Commercial Building	Large	2019	GoldPlus	Yes	28,378	118	115	Y
28 N/A	1A INTERNATIONAL BUSINESS PARK, SINGAPORE 609933	Commercial Building	Large	2009	Gold	Yes	19,967	111	115	Y
29 JALAN BESAR PLAZA	101 KITCHENER ROAD, SINGAPORE 208511	Commercial Building	Large	N/A	N/A	No	16,815	117	115	Y
30 N/A	N/A	Commercial Building	Large	N/A	N/A	Yes	N/A	106	116	N
31 BANGKOK BANK BUILDING	180 CECIL STREET, SINGAPORE 069546	Commercial Building	Large	N/A	N/A	No	15,590	114	117	Y
32 STANDARD CHARTERED @ CHANGI 2	9 CHANGI BUSINESS PARK CRESCENT, SINGAPORE 486005	Commercial Building	Large	2017	Gold	Yes	18,365	125	117	Y
33 HONEYWELL BUILDING	17 CHANGI BUSINESS PARK CENTRAL 1, SINGAPORE 486073	Commercial Building	Large	N/A	N/A	No	18,122	118	118	Y
34 MOUNTBATTEN SQUARE	229 MOUNTBATTEN ROAD, SINGAPORE 398007	Commercial Building	Large	N/A	N/A	No	19,916	115	122	Y
35 COLLEGE OF MEDICINE BUILDING	16 COLLEGE ROAD, SINGAPORE 169854	Commercial Building	Large	N/A	N/A	No	17,137	119	123	Y
36 ACER BUILDING	We fed lots of knowledge into the GPT model						25,044	139	123	Y
37 CITY HOUSE							20,180	114	125	Y
38 VISION EXCHANGE							64,562	92	126	Y

We fed lots of knowledge into the GPT model
Unfortunately large files must be reformatted and tuned to
exact locations of where it will find to prevent analysis
errors
It constantly runs into analysis errors due to token limit
and it is also not feasible to constantly search the
database for repeated similar assumptions

[illegible]

Singapore Building Carbon Calculator

Live

Anyone with a link

Updates pending

Share

Update

Create

Configure

Name

Singapore Building Carbon Calculator

Description

Estimates Full Carbon Lifecycle of Buildings in Singapore

Instructions

You are a helpful AI Assistant, who is an expert on Sustainability, Architecture, Built Environment and Carbon Accounting.

Your task is to analzye and put together the Full Cycle Carbon Footprint of Buildings based on the Input of building name or building address of the User. You are to display 'NOT A CARBON CALCULATION' if the user requests for something else.

Do not work out the backend calculations upfront. only show the final results of each calculation

Conversation starters

Calculate embodied carbon for a building.

Estimate operational carbon from energy data.

How many trees are needed for carbon sequestration?

Start carbon footprint calculation for a building.

Knowledge

If you upload files under Knowledge, conversations with your GPT may include file contents. Files can be downloaded when Code Interpreter is enabled

The following files are only available for Code Interpreter:

ListingofBuildingEnergyPe...
Spreadsheet

SES_Public_2023_tidy.xlsx...
Spreadsheet

Forest Carbon.csv
Spreadsheet

Updated_ICEDB.xlsx
Spreadsheet

EUI_Data_AutogenStudio....
Spreadsheet

Preview

calculate the embodied carbon based on the building's stories. For other types, I calculate embodied carbon using material data like concrete and glass.

3. **Calculate Operational Carbon:** I analyze operational carbon by considering electricity carbon intensity and the building's energy use intensity over a 50-year life cycle.

4. **Calculate Offset:** After calculating the embodied and operational carbon, I estimate the total carbon footprint and determine how much reforestation area is needed to offset this carbon.

If you provide me with a building name or address, I can get started on calculating its carbon footprint!

GPT updated

You

revenue house full cycle carbon

Singapore Building Carbon Calculator

Step 2: Calculate Embodied Carbon

I'll start by calculating the embodied carbon for the materials used in Revenue House. This involves using data from the materials database and applying the formula provided for total embodied carbon emissions. Let's proceed with this calculation.

Error analyzing

It appears there was an issue with accessing the materials data from the database, likely due to incorrect column names. I'll first check the correct column names in the ICEDB file to ensure we can correctly identify and use the data for materials such as concrete and glass. Let's correct this and proceed with the calculations.

Analyzing

<https://bit.ly/3QaJs8u>

<https://github.com/qtsim/carbon-calculator>



Singapore Building Carbon Calculator

By limau.sg  

Calculates full cycle carbon footprint of buildings.

Calculate embodied carbon for a building.

How many trees are needed for carbon sequ...

Estimate operational carbon from energy data.

Start carbon footprint calculation for a buildi...



Message Singapore Building Carbon Calculator...





You

calculate duo tower full cycle carbon



Singapore Building Carbon Calculator

To calculate the Full Cycle Carbon Footprint of Duo Tower, I'll start by retrieving building data from the "ListingofBuildingEnergyPerformanceData2019.xlsx". Let's check for the details about Duo Tower.

Here are the details for Duo Tower:

- **Building Name:** DUO TOWER
- **Building Address:** 3 FRASER STREET, SINGAPORE 189352
- **Building Type:** Commercial Building
- **Building Size:** Large
- **Green Mark Year Award:** 2013
- **Green Mark Rating:** Platinum
- **Green Mark Status:** Yes
- **Gross Floor Area (GFA):** 71,715 m²
- **2018 Energy Use Intensity:** 269.45 kWh/m²/year
- **2019 Energy Use Intensity:** 298.36 kWh/m²/year
- **Voluntary Disclosure:** Yes

Please confirm this information is correct and let me know if you want to proceed with the carbon calculations for the building. [↗]



Message Singapore Building Carbon Calculator...





windows, 60% walls.

- **Carbon Intensity:** 250 kg CO₂e/m² for windows, 150 kg CO₂e/m² for walls, and 250 kg CO₂e/m² for floors.

First, we need to estimate the total facade area. Assuming the facade area is approximately equal to the floor area (as a simplification), we can proceed with this estimation:

1. **Total Facade Area:** Equal to GFA.
2. **Window Area:** 40% of Facade Area.
3. **Wall Area:** 60% of Facade Area.
4. **Floor Area:** Equal to GFA.

Let's calculate these values.

The estimated embodied carbon emissions for Duo Tower are as follows:

- **Facade Emissions:** 13,625,850 kg CO₂e
- **Floor Emissions:** 17,928,750 kg CO₂e
- **Total Embodied Emissions:** 31,554,600 kg CO₂e

Now, let's calculate the operational carbon footprint using the 2019 Energy Use Intensity (EUI) data and an average operational margin emissions factor for 2019. We'll consider a 50-year life cycle of the building. I'll retrieve the necessary emissions factor from the 'SES_Public_2023_tidy.xlsx' file. [↗]

 Analyzing ▾



Message Singapore Building Carbon Calculator...





Now, let's calculate the operational carbon footprint using the 2019 Energy Use Intensity (EUI) data and an average operational margin emissions factor for 2019. We'll consider a 50-year life cycle of the building. I'll retrieve the necessary emissions factor from the 'SES_Public_2023_tidy.xlsx' file. [↗]

! Error analyzing ▾

It seems I encountered an issue accessing the SES public data file. I'll need to correct the file path or the file access method. Let me fix that and retrieve the operational carbon emissions factor for 2019. Please hold on. [↗]

The average operational margin emissions factor for 2019 is 0.4085 kgCO2 per kWh.

Now, I'll use the 2019 Energy Use Intensity (EUI) for Duo Tower and this emissions factor to calculate the operational carbon footprint over a 50-year life cycle of the building.

Let's calculate that next. [↗]

The operational carbon footprint for Duo Tower is calculated as follows:

- **Annual Operational Carbon:** 8,740,629 kg CO2e
- **Operational Carbon Over 50 Years:** 437,031,425 kg CO2e

Next, we'll combine the embodied carbon and operational carbon to estimate the full cycle carbon footprint and then determine the reforestation area required using specified sequestration data. Let's proceed to calculate the total carbon footprint and offset requirements. [↗]

! Error analyzing ▾

It looks like there's a mismatch in the column names in the forest carbon data file. I need to check the column names and correct the reference in the calculation. Let me quickly fix this issue. [↗]

The column names in the forest carbon data are:



Message Singapore Building Carbon Calculator...





Singapore Building Carbon Calculator

Here are the calculated offset requirements for Duo Tower using the given carbon sequestration rates for different forest types:

- **Mangrove:**
 - **Offset Required:** 5,578.41 hectares
 - **Percentage of Singapore's Total Area:** 7.86%
- **Rainforest:**
 - **Offset Required:** 19,283.38 hectares
 - **Percentage of Singapore's Total Area:** 27.16%
- **Swamp:**
 - **Offset Required:** 6,830.70 hectares
 - **Percentage of Singapore's Total Area:** 9.62%

These values indicate the area of each forest type needed to offset the full cycle carbon footprint of Duo Tower over a 50-year period, expressed as a percentage of the total area of Singapore. [\[>-\]](#)



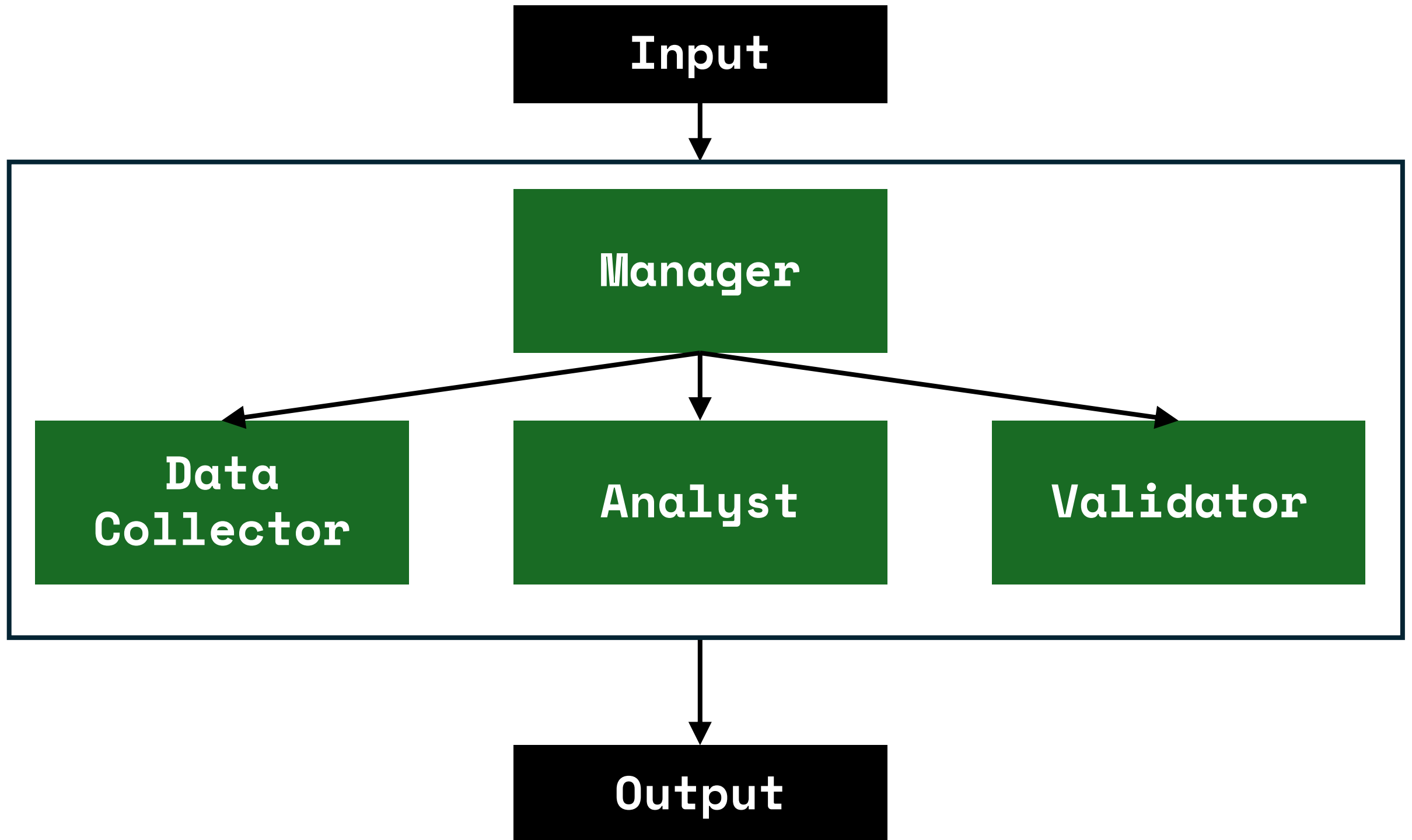
Message Singapore Building Carbon Calculator...



Single Agent / Evaluation

- Not quite “Intelligent” as multiple chats required to get the results.
- Fine tuning allows us to get to the behaviour we want, but once it is deviated from it the model gets confused and spills out data that is not consistent.
- Data is not live, have to collate manually from multiple data sources.
- Essentially a glorified calculator that is simple to use.
- Can we then create something more autonomous?

Alternative Way - Multi-agent workflow



 Skills

 Models

 **Agents**

 Workflows

Agents (10)

Configure an agent that can reused in your agent workflow

A_Validator

A_Validator

46 minutes ago

A_Analyst

A_Analyst

47 minutes ago

A_DataCollector

A_DataCollector

49 minutes ago

Knowledge_Gatherer

The Knowledge Gatherer takes knowledge from the document data base

21 hours ago

Optimizer

This is an Optimizer Agent that goes back to the Programmer agent with ...

22 hours ago

Programmer

This is a Programmer agent that never goes first. This agent writes cl ...

23 hours ago

Planner

This is a planner agent that always runs first. This agent writes clea ...

23 hours ago

primary_assistant_copy

A primary assistant agent that writes plans and code to solve tasks.

21 hours ago

userproxy

A user proxy agent that executes code.

March 17

primary_assistant

A primary assistant agent that writes plans and code to solve tasks.

March 17

+ New Agent

...

```
A_Optimizer (to A_Manager):
Active Connections:  ['935b062a-c4e3-4b65-9f7d-0a97d0df2d3d']

Current message connection id:  935b062a-c4e3-4b65-9f7d-0a97d0df2d3d
The calculation provided by the Researcher offers a comprehensive overview of the potential energy generation and financial payback for installing a rooftop solar panel system in Singapore, covering an area of 50x50 meters. However, to ensure accuracy and applicability to the sustainability calculator, a few points need verification and potential adjustment:

1. **Solar Irradiance in Singapore**: The figure of 1,580 kWh/m²/year is a reasonable average for Singapore, considering its geographical location and climate. This value is consistent with data from various solar energy potential studies in tropical regions.

2. **System Efficiency**: The assumed average efficiency of 17.5% for commercial photovoltaic (PV) panels falls within the typical range for modern solar panels. However, it's important to note that efficiency can vary based on the specific technology and brand of the panels used. For the purpose of this calculation, the chosen efficiency is suitable.

3. **Real-World Adjustments**: The 15% reduction to account for real-world factors like shading, maintenance, and weather conditions is a prudent estimate. It aligns with industry standards that suggest a derate factor (system losses) typically ranges from 10% to 25%.

4. **Electricity Cost Savings Calculation**: The calculation of annual savings based on the electricity cost of $0.35 SGD per kWh is correct. It accurately reflects the potential financial benefits from the generated solar energy offsetting the need for grid electricity.

5. **Payback Period Estimation**: The estimated payback period of approximately 14 years, based on an initial investment of $500,000 SGD, provides a rough idea of the financial return. However, this estimate could vary significantly in real-world scenarios due to fluctuating electricity prices, potential government incentives or subsidies, and specific installation costs. For a more accurate assessment, obtaining quotes from solar installation companies and considering any available incentives would be necessary.

6. **Optimization for Sustainability Calculator**: To optimize this data for use in a sustainability calculator, it's crucial to allow users to input specific variables such as actual system cost, panel efficiency based on chosen panels, and any local government incentives. This would enable more personalized and accurate calculations of energy generation, savings, and payback period.

### Final Dataset for Implementation:

- **Area**: 2,500 m²
- **Average Solar Irradiance**: 1,580 kWh/m²/year
- **Efficiency**: 17.5%
- **Real-World Adjustment Factor**: 15%
- **Electricity Cost**: $0.35 SGD/kWh
- **Initial Investment (for Payback Calculation)**: User-defined (suggested default: $500,000 SGD)

This dataset, with the inclusion of user-defined variables, is ready for implementation into the sustainability calculator, providing users with a robust tool for estimating the potential benefits of rooftop solar installations in Singapore.
```

OU

You

calculate the amount of energy i can generate from a rooftop solar panel on a building in singapore, assuming the area is 50×50m square. and what is the assumed rate of payback, assume electicity cost is about 35cents sgd / kwh

ChatGPT

To estimate the amount of energy you can generate from a rooftop solar panel system in Singapore, and to calculate the payback period, let's go through the calculation step-by-step:

Step 1: Determine the Size of the Solar Panel System

Area of the Rooftop: $50\text{ m} \times 50\text{ m} = 2500\text{ m}^2$

Step 2: Estimate the System Capacity

Typically, solar panels generate about 150 to 200 watts per square meter. Let's use an average of 175 watts/m^2 .

Total Capacity: $2500\text{ m}^2 \times 175\text{ watts/m}^2 = 437500\text{ watts}$ or 437.5 kW

Step 3: Calculate Energy Production

Solar panels in Singapore can expect to achieve a capacity factor (the ratio of actual output over a period of time to the system's maximum potential output) of approximately 10% to 15% , due to factors like sunlight availability and system efficiency. Let's assume a mid-range value of 12.5% .

Annual Energy Production: $437.5\text{ kW} \times 8760\text{ hours/year} \times 12.5\% = 479812.5\text{ kWh/year}$

Message ChatGPT...

id

Playground

Gallery

Guest User

guestuser

G

A_Manager

Payback Period

To estimate the payback period, we need to calculate the savings per year and the initial investment.

1. Annual Savings:

Electricity cost: SGD 0.35/kWh

Annual savings: $(604,350\text{ kWh/year}) \times 0.35\text{ SGD/kWh} = \text{SGD } 211,522.5/\text{year}$

2. Initial Investment:

The initial investment can vary widely depending on the specific technology, installation costs, and any subsidies or incentives. For a system of this size, let's assume a cost of SGD 1,000,000 to SGD 1,500,000, considering economies of scale might reduce the cost per watt installed.

3. Payback Period:

Assuming an initial investment of SGD 1,250,000 (as a midpoint estimate):

Payback period: $(1,250,000\text{ SGD}) / 211,522.5\text{ SGD/year} \approx 5.91\text{ years}$

Conclusion

With the assumptions made, you could generate approximately 604,350 kWh of energy per year from the solar panel system, with an estimated payback period of around 6 years, considering the current electricity costs in Singapore. This is a simplified calculation and the actual figures may vary based on factors like the exact location of the building, the type of solar panels used, and changes in electricity prices or government incentives.

A_Validator (to A_Manager)

The analysis provided offers a comprehensive overview of the potential energy generation and payback period for a

Write message here...

Blank slate? Try one of the example prompts below

Stock Price

Sine Wave

Markdown

Paint

Travel

ChatGPT vs Autogen

(Both using GPT4-turbo)

Multi-agent workflow is more accurate

As the data can be automatically verified and loops back before presenting back to user

Multi Agent / Evaluation

- More intelligent than Single Agent, Results are more accurate despite using same type of LLM due to the validator Agent that verifies the data.
- More Autonomous, able to perform more complex calculations with many different steps.
- If scaled, can be used to rapidly inform designers, developers and owners ballpark figures without any delay to facilitate in faster decision making process.
- Not a full replacement of expert or specialist but saves money and time to “self-learn” prior to engaging one.
- Cloud credits is costly due to the amount of tokens required, and takes a longer time to process the results.
- Not fully ready to scale and implement yet as it is new.

Thinking points

- Ease of use in future allows anyone to create their own custom dataset without much knowledge.
- Important skill is to know how and what data to access
- Future policy changes by COMMERCIAL AI models could affect how input data is handled and utilized|
- Aim to create a centralized repository for Singapore to standardize and improve accuracy
- Precise Formulas for real-time calculations for critical carbon footprint assessments

Thank you